

Abstract

We report a numerical correspondence between the Betti numbers of a specific G_2 -holonomy manifold and the experimentally measured leptonic CP-violating phase. The manifold X_7 is constructed via the Twisted Connected Sum method using a Nikulin involution on K3 surfaces. The mathematical structure of the Nikulin lattice $L_+ \cong E_8(-1) \oplus U \oplus U \oplus \langle -2 \rangle$ uniquely determines the Betti numbers $(b_2, b_3) = (13, 7)$. From these topological invariants, we obtain $\delta_{\text{CP}} = 2\pi(b_3^2 + 1)/(b_2 \cdot b_3) = 197.8$. This result involves no adjustable parameters. The current experimental value is $\delta_{\text{CP}} = 197 \pm 25$ (NuFit 6.0, normal ordering), consistent with the topological prediction at 0.03σ . If future measurements by DUNE determine δ_{CP} to lie outside the range 188–208, this correspondence is excluded.

Topological Invariants of G_2 -Manifolds and the Leptonic CP Phase: A Parameter-Free Prediction

TdP Collaboration
Red II Research Program

February 7, 2026

1 Introduction

The leptonic CP-violating phase δ_{CP} in the Pontecorvo-Maki-Nakagawa-Sakata (PMNS) neutrino mixing matrix remains among the least constrained parameters of the Standard Model. Current global fits yield $\delta_{\text{CP}} \approx 200$ with uncertainties of order 25 [1]. No theoretical framework predicts this value from first principles.

We report a numerical correspondence: the Betti numbers of a mathematically well-defined G_2 -holonomy manifold, when combined in a specific topological formula, reproduce the experimentally observed value of δ_{CP} .

We state precisely what this paper does and does not claim:

Claim: A specific topological construction, based on established mathematics, yields $\delta_{\text{CP}} = 197.8$, consistent with current experimental data.

Not claimed: That this construction is the unique compactification relevant to our universe, nor that the complete Standard Model emerges from it.

The value of this observation lies in its falsifiability. The Deep Underground Neutrino Experiment (DUNE) [2] is projected to measure δ_{CP} with precision of approximately 5 by 2030. If δ_{CP} is determined to lie outside the range 188–208, the correspondence reported here is excluded.

2 The Nikulin-Kovalev Construction

2.1 G_2 manifolds from Twisted Connected Sums

Compact 7-dimensional manifolds with G_2 holonomy can be constructed via the Twisted Connected Sum (TCS) method developed by Kovalev [3] and extended by Corti, Haskins, Nordström, and Pacini [4]. The construction glues two asymptotically cylindrical Calabi-Yau 3-folds $Z_{\pm}$ along a common asymptotic region:

$$X_7 = (Z_+ \times S^1) \cup_{\varphi} (Z_- \times S^1) \quad (1)$$

where the gluing map φ involves a hyper-Kähler rotation of the K3 surface S appearing at infinity. The topology of X_7 is determined by the choice of involution $\sigma : S \rightarrow S$.

2.2 The Nikulin lattice

We employ the Nikulin involution [5], a holomorphic involution on the K3 surface characterized by its action on the second cohomology. For the Nikulin involution with fixed locus consisting of 8 isolated points and one rational curve, the invariant sublattice of $H^2(S; \mathbb{Z})$ has the structure:

$$L_+ \cong E_8(-1) \oplus U \oplus U \oplus \langle -2 \rangle \quad (2)$$

The components are standard objects in lattice theory:

- $E_8(-1)$: the negative-definite E_8 root lattice (rank 8)
- U : the hyperbolic plane with Gram matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ (rank 2, two copies)
- $\langle -2 \rangle$: the rank-1 lattice generated by a vector of norm -2

The total rank is $8 + 2 + 2 + 1 = 13$. Since $\text{rank}(H^2(S; \mathbb{Z})) = 22$ for any K3 surface, the anti-invariant lattice has rank:

$$\text{rank}(L_-) = 22 - 13 = 9 \quad (3)$$

We emphasize that the structure (2) is uniquely determined by the classification of Nikulin involutions. The values 13 and 9 are mathematical consequences, not adjustable parameters.

2.3 Betti numbers of the G_2 manifold

The Betti numbers of the TCS manifold X_7 are computed via the Mayer-Vietoris sequence. Following [4]:

$$b_2(X_7) = \text{rank}(L_+) = 13 \quad (4)$$

$$b_3(X_7) = \text{rank}(L_-) - 2 = 7 \quad (5)$$

The correction “ -2 ” in Eq. (5) arises from two cohomology classes in L_- that become identified under the matching conditions of the TCS construction. The result is:

$$(b_2, b_3) = (13, 7) \quad (6)$$

3 Derivation of the CP Phase

3.1 Topological ansatz

In M-theory compactifications on G_2 -holonomy manifolds, CP-violating phases in the effective 4D theory arise from the integration of the M-theory 3-form potential over calibrated 3-cycles [6]. The complex phases of Yukawa couplings receive contributions from membrane instantons wrapping associative submanifolds.

We consider the following topological ansatz for the leptonic CP phase:

$$\delta_{\text{CP}} = \frac{2\pi \cdot N_{\text{CP}}}{k_{\text{CS}}} \quad (7)$$

where:

- $k_{\text{CS}} = b_2 \times b_3$ is an effective Chern-Simons level counting gauge-matter couplings in the dimensional reduction.
- $N_{\text{CP}} = b_3^2 + 1$ counts CP-odd topological configurations, with b_3^2 accounting for instanton-anti-instanton pairs and the $+1$ for the trivial sector.

3.2 Numerical evaluation

Substituting the Betti numbers (6):

$$k_{\text{CS}} = 13 \times 7 = 91 \quad (8)$$

$$N_{\text{CP}} = 7^2 + 1 = 50 \quad (9)$$

$$\delta_{\text{CP}} = \frac{2\pi \times 50}{91} = \frac{100\pi}{91} \quad (10)$$

Converting to degrees:

$$\boxed{\delta_{\text{CP}} = 197.80} \quad (11)$$

3.3 Parameter accounting

We emphasize the absence of adjustable parameters:

Category	Count
Continuous parameters	0
Discrete choices fitted to data	0

The values $b_2 = 13$ and $b_3 = 7$ are derived from the Nikulin lattice structure, not selected to match experimental data.

4 Experimental Comparison

4.1 Current status

The NuFit 6.0 global analysis [1] reports for normal neutrino mass ordering:

$$\delta_{\text{CP}}^{\text{exp}} = 197 \pm 25 \quad (1\sigma) \quad (12)$$

The prediction (11) deviates from the experimental central value by 0.8, corresponding to 0.03σ .

Source	δ_{CP}
This work	197.8
NuFit 6.0 (NO)	197 ± 25

Table 1: Comparison of the topological prediction with current experimental constraints.

4.2 Falsification criterion

The DUNE experiment is expected to achieve precision of approximately 5 on δ_{CP} by circa 2030 [2]. We define:

Falsification criterion: If DUNE determines $|\delta_{\text{CP}} - 197.8| > 10$, the correspondence is excluded at greater than 2σ confidence.

Equivalently, if $\delta_{\text{CP}}^{\text{DUNE}}$ falls outside the range $[188, 208]$, this framework is falsified. There is no mechanism for post-hoc adjustment; the prediction stands or falls with experiment.

5 Discussion

5.1 Status of the result

This work identifies a numerical correspondence between:

1. A topological invariant computed from established mathematical constructions (TCS, Nikulin lattice classification).
2. A physical observable measured in neutrino oscillation experiments.

We do not claim this correspondence is *explained*. The formula (7) is an ansatz motivated by instanton counting in M-theory, not a rigorous derivation from first principles. Whether the correspondence reflects underlying physics or is accidental is a question for experiment to resolve.

5.2 Limitations

This work concerns only the prediction of δ_{CP} . We do not derive fermion masses, gauge coupling constants, or other Standard Model parameters. The topological construction determines the Betti numbers but does not, by itself, specify the detailed particle content.

5.3 Speculative extensions

We note briefly that certain hadronic mass scales appear to follow geometric scaling relations when the golden ratio $\varphi = (1 + \sqrt{5})/2$ is introduced:

$$m_p \approx k_{\text{CS}} \times 2\pi \times \varphi \approx 925 \text{ MeV} \quad (13)$$

$$\Delta m_{n-p} \approx m_e \times \varphi^2 \approx 1.34 \text{ MeV} \quad (14)$$

These relations have deviations of 1–3% from experimental values. However, they require the emergence of φ from the optimal G_2 metric, which is conjectured but not established. Understanding this would require detailed knowledge of moduli stabilization in G_2 compactifications, which lies beyond the scope of this report.

The core prediction (11) is independent of these speculative extensions and depends only on the Betti numbers (13, 7).

6 Conclusion

We have reported a parameter-free numerical correspondence:

$$\delta_{\text{CP}} = \frac{2\pi(b_3^2 + 1)}{b_2 \cdot b_3} = 197.8$$

where $(b_2, b_3) = (13, 7)$ are the Betti numbers of a G_2 -holonomy manifold constructed via Twisted Connected Sum with Nikulin involution. These Betti numbers are determined by the lattice $E_8(-1) \oplus U \oplus U \langle -2 \rangle$, not fitted to experimental data.

The prediction is consistent with current measurements and will be tested definitively by DUNE. If δ_{CP} is measured outside the range 188–208, this correspondence is falsified.

Acknowledgments

We thank critical reviewers whose feedback led to the clear separation of the falsifiable core prediction from speculative extensions.

References

- [1] I. Esteban, M. C. Gonzalez-Garcia, M. Maltoni, T. Schwetz, and A. Zhou,

- “The fate of hints: updated global analysis of three-flavor neutrino oscillations,” JHEP **09**, 178 (2020); updated results available at www.nu-fit.org (2024).
- [2] DUNE Collaboration, “Long-Baseline Neutrino Facility (LBNF) and Deep Underground Neutrino Experiment (DUNE) Conceptual Design Report,” arXiv:1601.05471 (2016).
 - [3] A. Kovalev, “Twisted connected sums and special Riemannian holonomy,” J. Reine Angew. Math. **565**, 125 (2003).
 - [4] A. Corti, M. Haskins, J. Nordström, and T. Pacini, “ G_2 -manifolds and associative submanifolds via semi-Fano 3-folds,” Duke Math. J. **164**, 1971 (2015).
 - [5] V. V. Nikulin, “Integer symmetric bilinear forms and some of their geometric applications,” Izv. Akad. Nauk SSSR Ser. Mat. **43**, 111 (1979).
 - [6] E. Witten, “On flux quantization in M-theory and the effective action,” J. Geom. Phys. **22**, 1 (1997).